

FINITELY PRESENTED GROUPS

Although finitely generated [FREE GROUPS](#) and [ONE-RELATOR GROUPS](#) are finitely presented, we believe they deserve special sections, so you won't find them here.

(FP1) (W.Magnus) The triviality problem for groups with a balanced presentation (the number of generators equals the number of relators). See also problem (O1).

(FP2) Can a non-trivial finitely presented group be isomorphic to its direct square? [Background](#)

*(FP3) (R.Bieri, R.Strebel) Is it true that if the relation module of a group G is finitely generated, then G is finitely presented? [Background](#)

*(FP4) (J.Stallings) If a finitely presented group is trivial, is it always possible to replace one of the defining relators by a primitive element without changing the group? [Background](#)

(FP5) (C.Y.Tang) Is there a non-free non-cyclic finitely presented group all of whose proper subgroups are free?

(FP6) Is every knot group virtually free-by-cyclic?

*(FP7) (G.Baumslag) Is every finitely generated group discriminated by a free group, finitely presented? [Background](#)

*(FP8) (G.Baumslag) Is a finitely generated free-by-cyclic group finitely presented? [Background](#)

(FP9) (G.Baumslag, F.B.Cannonito, C.F.Miller) Is every countable locally linear group embeddable in a finitely presented group?

(FP10) (A.Olshanskii) If a relatively free group is finitely presented, is it virtually nilpotent?

(FP11) (S.Ivanov) Is every finitely presented Noetherian group virtually polycyclic?

(FP12) (M.I.Kargapolov) Is every residually finite Noetherian group virtually polycyclic?

(FP13) (J.Wiegold) Is every finitely generated perfect group G (i.e., $[G, G] = G$) the normal closure of a single element?

*(FP14) (P.Scott) Let p, q, r be distinct prime numbers. Is the free product $Z_p * Z_q * Z_r$ the normal closure of a single element? [Background](#)

*(FP15) (B.Fine) Let G be an n -generator group. Call a set of elements $\{g_1, \dots, g_k\}$, $k \leq n$, a *test set* for the group G if, whenever $f(g_i) = g_i$, $i = 1, \dots, k$, for some endomorphism f of the group G , this f is actually an automorphism of G . The *test rank* of G is the minimal cardinality of a test set. Can the test rank be equal to 2 if $n > 2$? [Background](#)

(FP16) (D.Anosov) Is there a non-cyclic finitely presented group each element of which is a conjugate of some power of a single element?

(FP17) (V.N.Remeslennikov) Is every countable abelian group embeddable in the centre of some finitely presented group?

*(FP18) (R.Hirshon) Let G be a finitely generated residually finite group, and f an endomorphism of G . Is it true that $f^{\wedge\{k+1\}}(G)$ is isomorphic to $f^{\wedge k}(G)$ for some k ? [Background](#)

(FP19) (J.Makowsky) Is there an infinite finitely presented group with finitely many conjugacy classes? [Background](#)

* (FP20) (V.Guba) Is there a finitely generated group, other than $\mathbb{Z}_2$, with exactly 2 conjugacy classes? [Background](#)

(FP21) (E.Zelmanov) Let F_n be the free group of rank n , and P_m the subgroup of F_n generated by m th powers of all *primitive elements* of F_n . (This subgroup is obviously normal in F_n). Is it true that the factor group $BP(n, m) = F_n/P_m$ is NOT residually finite for sufficiently large m ?

(FP22) (R.I.Grigorchuk) Is it true that every finitely presented group contains either a free 2-generator semigroup, or a nilpotent subgroup of finite index?

(FP23) (R.I.Grigorchuk) Let F be Thompson's group:
 $(x_0, x_1, \dots \mid x_i x_k x_i^{-1} = x_{k+1}, \quad k > i, \quad k=1,2,\dots)$.
 Is F amenable? [Background](#)

(FP24) (V.Guba) Let $k[F]$ be the group ring of Thompson's group F (see Problem (FP23)) over a field k . Does $k[F]$ satisfy the Ore condition? That is, for any $a, b \in k[F]$, are there nonzero $u, v \in k[F]$ such that $au = bv$? (If the answer is negative, then F is not amenable.)